MSG No. 209.48 $F432$ [Type I, cubic]

Table 1: Wyckoff site: 4a, site symmetry: 432

No.	position	mapping
1	$[0, 0, 0]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72]
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96]

Table 2: Wyckoff site: 4b, site symmetry: 432

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[\frac{1}{2}, 0, 0\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]
3	$\left[0, \frac{1}{2}, 0\right]$	[49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72]
4	$\left[0, 0, \frac{1}{2}\right]$	[73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96]

Table 3: Wyckoff site: 8c, site symmetry: 23.

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,17,18,32,46,47,57,67,72,82,92,93]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,7,15,27,28,35,60,62,64,77,78,85]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,4,11,26,31,39,53,54,61,84,86,88]
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5,6,13,36,38,40,51,52,59,74,79,87]
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[8,22,23,25,41,42,58,68,69,81,91,96]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[9,19,24,34,44,45,49,65,66,80,94,95]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[10,20,21,33,43,48,56,70,71,73,89,90]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[12,14,16,29,30,37,50,55,63,75,76,83]

Table 4: Wyckoff site: 24d, site symmetry: 2.22

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[1,13,32,38]
2	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[2,10,27,33]
3	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[3,9,26,34]
4	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[4,18,88,93]
5	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[5,19,87,95]
6	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[6,20,59,70]
7	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[7,17,60,72]
8	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[8,14,25,37]
9	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[11,22,54,68]
10	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[12,24,55,65]
11	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[15,23,77,91]
12	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[16,21,76,90]
13	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[28,42,64,69]

continued ...

Table 4

No.	position	mapping
14	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[29,43,63,71]
15	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[30,44,83,94]
16	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[31,41,84,96]
17	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[35,46,78,92]
18	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[36,48,79,89]
19	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[39,47,53,67]
20	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[40,45,52,66]
21	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[49,61,80,86]
22	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[50,58,75,81]
23	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[51,57,74,82]
24	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[56,62,73,85]

Table 5: Wyckoff site: 24e, site symmetry: 4..

No.	position	mapping
1	$[x, 0, 0]$	[1,2,3,8]
2	$[0, 0, -x]$	[4,16,19,23]
3	$[0, 0, x]$	[5,15,18,21]
4	$[0, x, 0]$	[6,11,17,24]
5	$[0, -x, 0]$	[7,12,20,22]
6	$[-x, 0, 0]$	[9,10,13,14]
7	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,32]
8	$\left[0, \frac{1}{2}, \frac{1}{2} - x\right]$	[28,40,43,47]
9	$\left[0, \frac{1}{2}, x + \frac{1}{2}\right]$	[29,39,42,45]
10	$\left[0, x + \frac{1}{2}, \frac{1}{2}\right]$	[30,35,41,48]
11	$\left[0, \frac{1}{2} - x, \frac{1}{2}\right]$	[31,36,44,46]
12	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[33,34,37,38]
13	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	[49,50,51,56]
14	$\left[\frac{1}{2}, 0, \frac{1}{2} - x\right]$	[52,64,67,71]
15	$\left[\frac{1}{2}, 0, x + \frac{1}{2}\right]$	[53,63,66,69]
16	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[54,59,65,72]
17	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[55,60,68,70]
18	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	[57,58,61,62]
19	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	[73,74,75,80]
20	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[76,88,91,95]
21	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[77,87,90,93]
22	$\left[\frac{1}{2}, x + \frac{1}{2}, 0\right]$	[78,83,89,96]
23	$\left[\frac{1}{2}, \frac{1}{2} - x, 0\right]$	[79,84,92,94]
24	$\left[\frac{1}{2} - x, \frac{1}{2}, 0\right]$	[81,82,85,86]

Table 6: Wyckoff site: 32f, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,17,18]

continued ...

Table 6

No.	position	mapping
2	$[x, -x, x]$	[2,7,15]
3	$[x, x, -x]$	[3,4,11]
4	$[-x, x, x]$	[5,6,13]
5	$[x, -x, -x]$	[8,22,23]
6	$[-x, x, -x]$	[9,19,24]
7	$[-x, -x, x]$	[10,20,21]
8	$[-x, -x, -x]$	[12,14,16]
9	$\left[x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[25,41,42]
10	$\left[x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[26,31,39]
11	$\left[x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[27,28,35]
12	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[29,30,37]
13	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[32,46,47]
14	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[33,43,48]
15	$\left[-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[34,44,45]
16	$\left[-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[36,38,40]
17	$\left[x+\frac{1}{2}, x, x+\frac{1}{2}\right]$	[49,65,66]
18	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[50,55,63]
19	$\left[x+\frac{1}{2}, x, \frac{1}{2}-x\right]$	[51,52,59]
20	$\left[\frac{1}{2}-x, x, x+\frac{1}{2}\right]$	[53,54,61]
21	$\left[x+\frac{1}{2}, -x, \frac{1}{2}-x\right]$	[56,70,71]
22	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[57,67,72]
23	$\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$	[58,68,69]
24	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-x\right]$	[60,62,64]
25	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x\right]$	[73,89,90]
26	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x\right]$	[74,79,87]
27	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[75,76,83]
28	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x\right]$	[77,78,85]
29	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$	[80,94,95]
30	$\left[\frac{1}{2}-x, x+\frac{1}{2}, -x\right]$	[81,91,96]
31	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[82,92,93]
32	$\left[\frac{1}{2}-x, \frac{1}{2}-x, -x\right]$	[84,86,88]

Table 7: Wyckoff site: 48g, site symmetry: ..2

No.	position	mapping
1	$[0, y, y]$	[1,13]
2	$[0, -y, y]$	[2,10]
3	$[0, y, -y]$	[3,9]
4	$[y, y, 0]$	[4,18]
5	$[-y, y, 0]$	[5,19]
6	$[-y, 0, y]$	[6,20]
7	$[y, 0, y]$	[7,17]
8	$[0, -y, -y]$	[8,14]
9	$[y, 0, -y]$	[11,22]
10	$[-y, 0, -y]$	[12,24]

continued ...

Table 7

No.	position	mapping
11	$[y, -y, 0]$	[15,23]
12	$[-y, -y, 0]$	[16,21]
13	$\left[0, y+\frac{1}{2}, y+\frac{1}{2}\right]$	[25,37]
14	$\left[0, \frac{1}{2}-y, y+\frac{1}{2}\right]$	[26,34]
15	$\left[0, y+\frac{1}{2}, \frac{1}{2}-y\right]$	[27,33]
16	$\left[y, y+\frac{1}{2}, \frac{1}{2}\right]$	[28,42]
17	$\left[-y, y+\frac{1}{2}, \frac{1}{2}\right]$	[29,43]
18	$\left[-y, \frac{1}{2}, y+\frac{1}{2}\right]$	[30,44]
19	$\left[y, \frac{1}{2}, y+\frac{1}{2}\right]$	[31,41]
20	$\left[0, \frac{1}{2}-y, \frac{1}{2}-y\right]$	[32,38]
21	$\left[y, \frac{1}{2}, \frac{1}{2}-y\right]$	[35,46]
22	$\left[-y, \frac{1}{2}, \frac{1}{2}-y\right]$	[36,48]
23	$\left[y, \frac{1}{2}-y, \frac{1}{2}\right]$	[39,47]
24	$\left[-y, \frac{1}{2}-y, \frac{1}{2}\right]$	[40,45]
25	$\left[\frac{1}{2}, y, y+\frac{1}{2}\right]$	[49,61]
26	$\left[\frac{1}{2}, -y, y+\frac{1}{2}\right]$	[50,58]
27	$\left[\frac{1}{2}, y, \frac{1}{2}-y\right]$	[51,57]
28	$\left[y+\frac{1}{2}, y, \frac{1}{2}\right]$	[52,66]
29	$\left[\frac{1}{2}-y, y, \frac{1}{2}\right]$	[53,67]
30	$\left[\frac{1}{2}-y, 0, y+\frac{1}{2}\right]$	[54,68]
31	$\left[y+\frac{1}{2}, 0, y+\frac{1}{2}\right]$	[55,65]
32	$\left[\frac{1}{2}, -y, \frac{1}{2}-y\right]$	[56,62]
33	$\left[y+\frac{1}{2}, 0, \frac{1}{2}-y\right]$	[59,70]
34	$\left[\frac{1}{2}-y, 0, \frac{1}{2}-y\right]$	[60,72]
35	$\left[y+\frac{1}{2}, -y, \frac{1}{2}\right]$	[63,71]
36	$\left[\frac{1}{2}-y, -y, \frac{1}{2}\right]$	[64,69]
37	$\left[\frac{1}{2}, y+\frac{1}{2}, y\right]$	[73,85]
38	$\left[\frac{1}{2}, \frac{1}{2}-y, y\right]$	[74,82]
39	$\left[\frac{1}{2}, y+\frac{1}{2}, -y\right]$	[75,81]
40	$\left[y+\frac{1}{2}, y+\frac{1}{2}, 0\right]$	[76,90]
41	$\left[\frac{1}{2}-y, y+\frac{1}{2}, 0\right]$	[77,91]
42	$\left[\frac{1}{2}-y, \frac{1}{2}, y\right]$	[78,92]
43	$\left[y+\frac{1}{2}, \frac{1}{2}, y\right]$	[79,89]
44	$\left[\frac{1}{2}, \frac{1}{2}-y, -y\right]$	[80,86]
45	$\left[y+\frac{1}{2}, \frac{1}{2}, -y\right]$	[83,94]
46	$\left[\frac{1}{2}-y, \frac{1}{2}, -y\right]$	[84,96]
47	$\left[y+\frac{1}{2}, \frac{1}{2}-y, 0\right]$	[87,95]
48	$\left[\frac{1}{2}-y, \frac{1}{2}-y, 0\right]$	[88,93]

Table 8: Wyckoff site: 48h, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{2}, y, y\right]$	[1,13]
2	$\left[\frac{1}{2}, -y, y\right]$	[2,10]
3	$\left[\frac{1}{2}, y, -y\right]$	[3,9]

continued ...

Table 8

No.	position	mapping
4	$\left[y,\, y,\, \frac{1}{2}\right]$	[4,18]
5	$\left[-y,\, y,\, \frac{1}{2}\right]$	[5,19]
6	$\left[-y,\, \frac{1}{2},\, y\right]$	[6,20]
7	$\left[y,\, \frac{1}{2},\, y\right]$	[7,17]
8	$\left[\frac{1}{2},\, -y,\, -y\right]$	[8,14]
9	$\left[y,\, \frac{1}{2},\, -y\right]$	[11,22]
10	$\left[-y,\, \frac{1}{2},\, -y\right]$	[12,24]
11	$\left[y,\, -y,\, \frac{1}{2}\right]$	[15,23]
12	$\left[-y,\, -y,\, \frac{1}{2}\right]$	[16,21]
13	$\left[\frac{1}{2},\, y+\frac{1}{2},\, y+\frac{1}{2}\right]$	[25,37]
14	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, y+\frac{1}{2}\right]$	[26,34]
15	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-y\right]$	[27,33]
16	$\left[y,\, y+\frac{1}{2},\, 0\right]$	[28,42]
17	$\left[-y,\, y+\frac{1}{2},\, 0\right]$	[29,43]
18	$\left[-y,\, 0,\, y+\frac{1}{2}\right]$	[30,44]
19	$\left[y,\, 0,\, y+\frac{1}{2}\right]$	[31,41]
20	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-y\right]$	[32,38]
21	$\left[y,\, 0,\, \frac{1}{2}-y\right]$	[35,46]
22	$\left[-y,\, 0,\, \frac{1}{2}-y\right]$	[36,48]
23	$\left[y,\, \frac{1}{2}-y,\, 0\right]$	[39,47]
24	$\left[-y,\, \frac{1}{2}-y,\, 0\right]$	[40,45]
25	$\left[0,\, y,\, y+\frac{1}{2}\right]$	[49,61]
26	$\left[0,\, -y,\, y+\frac{1}{2}\right]$	[50,58]
27	$\left[0,\, y,\, \frac{1}{2}-y\right]$	[51,57]
28	$\left[y+\frac{1}{2},\, y,\, 0\right]$	[52,66]
29	$\left[\frac{1}{2}-y,\, y,\, 0\right]$	[53,67]
30	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, y+\frac{1}{2}\right]$	[54,68]
31	$\left[y+\frac{1}{2},\, \frac{1}{2},\, y+\frac{1}{2}\right]$	[55,65]
32	$\left[0,\, -y,\, \frac{1}{2}-y\right]$	[56,62]
33	$\left[y+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-y\right]$	[59,70]
34	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, \frac{1}{2}-y\right]$	[60,72]
35	$\left[y+\frac{1}{2},\, -y,\, 0\right]$	[63,71]
36	$\left[\frac{1}{2}-y,\, -y,\, 0\right]$	[64,69]
37	$\left[0,\, y+\frac{1}{2},\, y\right]$	[73,85]
38	$\left[0,\, \frac{1}{2}-y,\, y\right]$	[74,82]
39	$\left[0,\, y+\frac{1}{2},\, -y\right]$	[75,81]
40	$\left[y+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[76,90]
41	$\left[\frac{1}{2}-y,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[77,91]
42	$\left[\frac{1}{2}-y,\, 0,\, y\right]$	[78,92]
43	$\left[y+\frac{1}{2},\, 0,\, y\right]$	[79,89]
44	$\left[0,\, \frac{1}{2}-y,\, -y\right]$	[80,86]
45	$\left[y+\frac{1}{2},\, 0,\, -y\right]$	[83,94]
46	$\left[\frac{1}{2}-y,\, 0,\, -y\right]$	[84,96]
47	$\left[y+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[87,95]
48	$\left[\frac{1}{2}-y,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[88,93]

Table 9: Wyckoff site: `48i`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,32]`
2	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,27]`
3	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,26]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ -x\right]$	`[4,88]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	`[5,87]`
6	$\left[\frac{3}{4},\ x,\ \frac{1}{4}\right]$	`[6,59]`
7	$\left[\frac{1}{4},\ -x,\ \frac{1}{4}\right]$	`[7,60]`
8	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[8,25]`
9	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[9,34]`
10	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10,33]`
11	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[11,54]`
12	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[12,55]`
13	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,38]`
14	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[14,37]`
15	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	`[15,77]`
16	$\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$	`[16,76]`
17	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[17,72]`
18	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	`[18,93]`
19	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	`[19,95]`
20	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[20,70]`
21	$\left[\frac{3}{4},\ \frac{3}{4},\ x\right]$	`[21,90]`
22	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	`[22,68]`
23	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	`[23,91]`
24	$\left[\frac{3}{4},\ x,\ \frac{3}{4}\right]$	`[24,65]`
25	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	`[28,64]`
26	$\left[\frac{3}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[29,63]`
27	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[30,83]`
28	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[31,84]`
29	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[35,78]`
30	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[36,79]`
31	$\left[\frac{1}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	`[39,53]`
32	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[40,52]`
33	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[41,96]`
34	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[42,69]`
35	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	`[43,71]`
36	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[44,94]`
37	$\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	`[45,66]`
38	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[46,92]`
39	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[47,67]`
40	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[48,89]`
41	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[49,80]`
42	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[50,75]`
43	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[51,74]`
44	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[56,73]`
45	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[57,82]`
46	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[58,81]`

continued ...

Table 9

No.	position	mapping
47	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[61,86]
48	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[62,85]

Table 10: Wyckoff site: 96j, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -z,\ y]$	[2]
3	$[x,\ z,\ -y]$	[3]
4	$[z,\ y,\ -x]$	[4]
5	$[-z,\ y,\ x]$	[5]
6	$[-y,\ x,\ z]$	[6]
7	$[y,\ -x,\ z]$	[7]
8	$[x,\ -y,\ -z]$	[8]
9	$[-x,\ y,\ -z]$	[9]
10	$[-x,\ -y,\ z]$	[10]
11	$[y,\ x,\ -z]$	[11]
12	$[-y,\ -x,\ -z]$	[12]
13	$[-x,\ z,\ y]$	[13]
14	$[-x,\ -z,\ -y]$	[14]
15	$[z,\ -y,\ x]$	[15]
16	$[-z,\ -y,\ -x]$	[16]
17	$[z,\ x,\ y]$	[17]
18	$[y,\ z,\ x]$	[18]
19	$[-y,\ z,\ -x]$	[19]
20	$[-z,\ -x,\ y]$	[20]
21	$[-y,\ -z,\ x]$	[21]
22	$[z,\ -x,\ -y]$	[22]
23	$[y,\ -z,\ -x]$	[23]
24	$[-z,\ x,\ -y]$	[24]
25	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[25]
26	$\left[x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	[26]
27	$\left[x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	[27]
28	$\left[z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	[28]
29	$\left[-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[29]
30	$\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[30]
31	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[31]
32	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[32]
33	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[33]
34	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[34]
35	$\left[y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[35]
36	$\left[-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[36]
37	$\left[-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[37]
38	$\left[-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[38]
39	$\left[z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	[39]

continued ...

Table 10

No.	position	mapping
40	$\left[-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	[40]
41	$\left[z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[41]
42	$\left[y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[42]
43	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	[43]
44	$\left[-z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	[44]
45	$\left[-y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	[45]
46	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[46]
47	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[47]
48	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[48]
49	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[49]
50	$\left[x+\frac{1}{2},\ -z,\ y+\frac{1}{2}\right]$	[50]
51	$\left[x+\frac{1}{2},\ z,\ \frac{1}{2}-y\right]$	[51]
52	$\left[z+\frac{1}{2},\ y,\ \frac{1}{2}-x\right]$	[52]
53	$\left[\frac{1}{2}-z,\ y,\ x+\frac{1}{2}\right]$	[53]
54	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	[54]
55	$\left[y+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[55]
56	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[56]
57	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[57]
58	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[58]
59	$\left[y+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[59]
60	$\left[\frac{1}{2}-y,\ -x,\ \frac{1}{2}-z\right]$	[60]
61	$\left[\frac{1}{2}-x,\ z,\ y+\frac{1}{2}\right]$	[61]
62	$\left[\frac{1}{2}-x,\ -z,\ \frac{1}{2}-y\right]$	[62]
63	$\left[z+\frac{1}{2},\ -y,\ x+\frac{1}{2}\right]$	[63]
64	$\left[\frac{1}{2}-z,\ -y,\ \frac{1}{2}-x\right]$	[64]
65	$\left[z+\frac{1}{2},\ x,\ y+\frac{1}{2}\right]$	[65]
66	$\left[y+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[66]
67	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[67]
68	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[68]
69	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[69]
70	$\left[z+\frac{1}{2},\ -x,\ \frac{1}{2}-y\right]$	[70]
71	$\left[y+\frac{1}{2},\ -z,\ \frac{1}{2}-x\right]$	[71]
72	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[72]
73	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[73]
74	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ y\right]$	[74]
75	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ -y\right]$	[75]
76	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ -x\right]$	[76]
77	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ x\right]$	[77]
78	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	[78]
79	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[79]
80	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[80]
81	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[81]
82	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[82]
83	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[83]
84	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	[84]
85	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ y\right]$	[85]
86	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ -y\right]$	[86]

continued ...

Table 10

No.	position	mapping
87	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, x\right]$	[87]
88	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, -x\right]$	[88]
89	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y\right]$	[89]
90	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x\right]$	[90]
91	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, -x\right]$	[91]
92	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y\right]$	[92]
93	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x\right]$	[93]
94	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, -y\right]$	[94]
95	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, -x\right]$	[95]
96	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, -y\right]$	[96]